

Exponential Distribution

Monte Carlo Simulation and Convergence Demonstrations

Sai Sampath Kedari

sampath@umich.edu

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1 Exponential Distribution

The Exponential distribution is the simplest continuous one-parameter model on $x > 0$ and an essential testbed for Monte Carlo simulation. It is defined by

$$P_X(x) = \begin{cases} \frac{1}{\lambda} e^{-\frac{x}{\lambda}} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where the scale parameter $\lambda > 0$ is equal to the true mean and the standard deviation:
 $\mathbb{E}[X] = \lambda, \text{Var}(X) = \lambda^2$

2 Random Variable Generation

To generate samples from the Exponential distribution, we use the Inverse Transform Sampling method.

Given a continuous random variable X with cumulative distribution function $F_X(x)$, if $U \sim U(0, 1)$, then

$$X = F_X^{-1}(U)$$

has the same distribution as X .

2.1 Derivation of the Inverse Transform

For the exponential density

$$P_X(x) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}, \quad \text{if } x \geq 0$$

the cumulative distribution function is

$$\begin{aligned} F_X(x) &= \int_0^x \frac{1}{\lambda} e^{-\frac{t}{\lambda}} dt \\ &= \frac{1}{\lambda} \int_0^x e^{-\frac{t}{\lambda}} dt \\ &= \frac{1}{\lambda} \cdot -\lambda \cdot e^{-\frac{t}{\lambda}} \Big|_0^x \\ &= 1 - e^{-\frac{x}{\lambda}} \end{aligned}$$

Let $U = F_X(x)$. Then

$$U = 1 - e^{-\frac{x}{\lambda}} \implies x = -\lambda \ln(1 - U).$$

Hence, the random variable generation formula is as follows.

$$X = -\lambda \ln(1 - U), \quad U \sim U(0, 1).$$

Since $1 - U$ is also uniform on $(0, 1)$, we commonly write

$$X = -\lambda \ln(U)$$

2.2 Inverse Transform Algorithm

To generate n i.i.d. samples from $\text{Exp}(\lambda)$:

1. Generate $U_i \sim U(0, 1)$ for $i = 1, \dots, n$.
2. Compute $X_i = -\lambda \ln U_i$.

Then $\{X_i\}_{i=1}^n$ are the i.i.d. drawn from the exponential distribution with the scale parameter λ . This exact inverse-transform relationship makes the Exponential distribution ideal for validating Monte Carlo estimators and convergence results in later sections.

2.3 Monte Carlo Verification

We generate samples using the inverse-transform function and compare the simulated distribution with the theoretical one to verify correctness.

```
def sample_exponential(lambda_param: float, n: int) -> np.ndarray:
    uniform_samples = np.random.rand(n)    #  $U \sim \text{Unif}(0, 1)$ 
    return -lambda_param * np.log(uniform_samples)
```

```
num_samples = 10000
lambda_parameter = 2
exponential_samples = sample_exponential(lambda_parameter, num_samples)
```

The histogram of the simulated samples aligns closely with the theoretical probability density function, confirming that the inverse-transform sampling is implemented correctly.

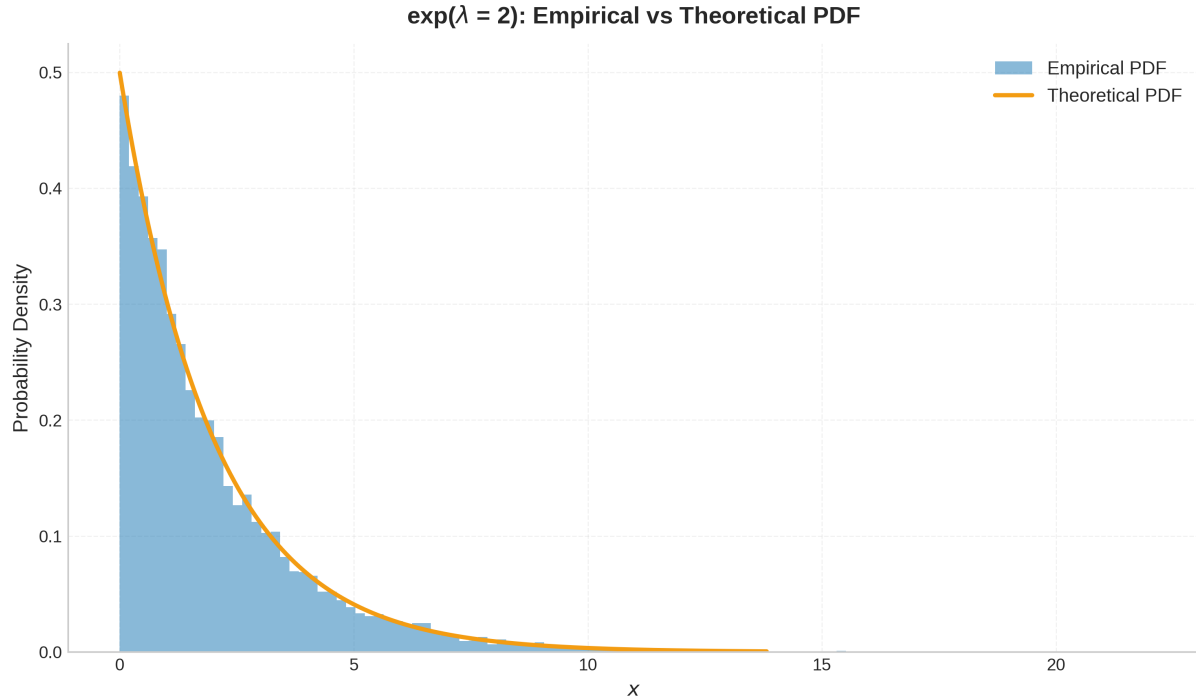


Figure 1: Empirical density of simulated Exponential samples ($n = 10^4$, $\lambda = 2$) compared with the theoretical pdf.

3 Convergence Demonstrations

The Law of Large Numbers (LLN) states that as the number of trials in a random experiment increases, the sample mean (average) of the outcomes approaches the expected value of the random variable.

In mathematical terms, if $X_1, X_2, X_3, \dots, X_n \stackrel{\text{iid}}{\sim} X$, where X is a random variable with a finite expected value $\mathbb{E}[X] < \infty$, then the sample mean,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Converges in probability to $\mathbb{E}[X]$ as n approaches infinity. This convergence implies that the larger the sample size n , the closer $\bar{X}_n$ will be to $\mathbb{E}[X]$, giving $\mathbb{E}[X]$ an operational meaning as the long-term average of the random variable X .

3.1 Weak Law of Large Numbers (WLLN)

The Weak Law of Large Numbers states that as the sample size n increases, the sample mean $\bar{X}_n$ becomes increasingly concentrated around the true mean $\mathbb{E}[X]$.

Let $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} X$ with $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2 < \infty$. Define the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then,

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}[\bar{X}_n] = \frac{\sigma^2}{n}.$$

As $n \rightarrow \infty$, the variance $\text{Var}[\bar{X}_n]$ approaches zero, implying that the distribution of $\bar{X}_n$ becomes more and more concentrated around its mean. Formally,

$$\bar{X}_n \xrightarrow[n \rightarrow \infty]{p} \mu,$$

that is, for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| > \epsilon) = 0.$$

Intuitively, if we repeat the same experiment many times, each time drawing n samples and computing their mean, we obtain a distribution of sample means. For small n , this distribution is wide and variable; for large n , it reaches a sharp peak around $\mathbb{E}[X]$. In the limit as $n \rightarrow \infty$, this sampling distribution collapses into a *constant (degenerate) distribution* at $\mathbb{E}[X]$, meaning that all probability mass accumulates at the true mean.

Hence, $\bar{X}_n$ converges *in probability* to $\mathbb{E}[X]$, and the sample mean behaves increasingly like a deterministic quantity as n grows.

```
def sampling_distribution(nsamples, ntrials, sample_generator, g_evaluator)
    Estimates = np.zeros((ntrials))
    for i in range(ntrials):
```

```

mc = monte_carlo(nsamples, sample_generator, g_evaluator)
Estimates[i] = mc.estimate
return Estimates

nsamples = np.array([[10, 100], [1000, 10000]])
ntrials = 1000

```

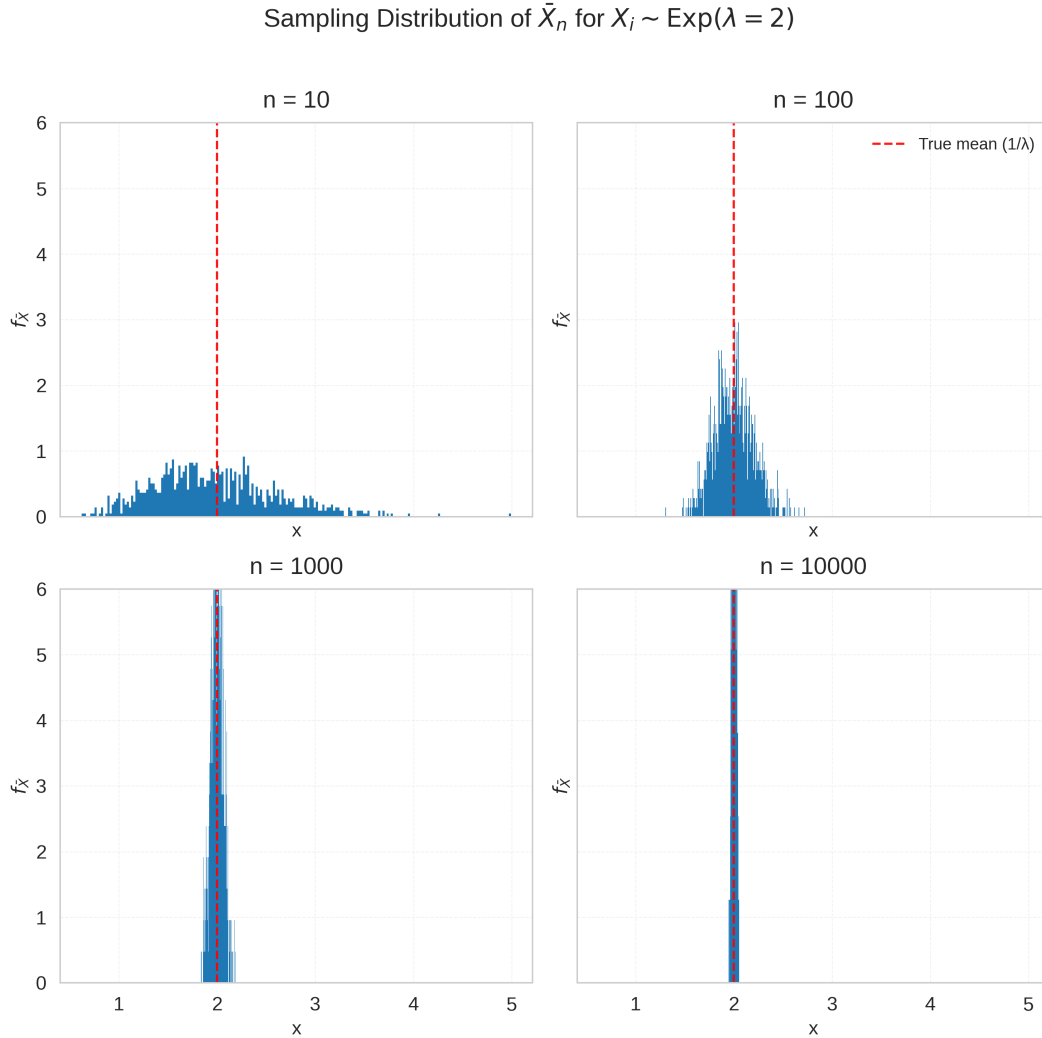


Figure 2: Sampling distribution of $\bar{X}_n$ for $X_i \sim \text{Exp}(\lambda = 2)$. As n increases, the distribution of $\bar{X}_n$ becomes increasingly concentrated around the true mean $\mathbb{E}[X] = \lambda = 2$, demonstrating convergence in probability.

3.2 Strong Law of Large Numbers (SLLN)

The Strong Law of Large Numbers strengthens the result of the Weak Law by asserting a more powerful mode of convergence: the sample mean $\bar{X}_n$ converges to the true mean $\mathbb{E}[X]$ *almost surely* (a.s.), that is, with probability one.

Let $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} X$ with $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2 < \infty$. Define the running sample

mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, by the Strong Law of Large Numbers,

$$\bar{X}_n \xrightarrow[n \rightarrow \infty]{a.s.} \mu,$$

that is, for every $\epsilon > 0$,

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \left| \bar{X}_n - \mathbb{E}[X] \right| > \epsilon\right) = 0, \quad \forall \epsilon > 0.$$

In words, the probability that the running average $\bar{X}_n$ ever deviates from the true mean $\mathbb{E}[X]$ by more than ϵ after some point n becomes zero as $n \rightarrow \infty$. Equivalently, the probability that a sequence *fails* to converge to the true mean is zero, or said another way, the probability that it *does* converge to $\mathbb{E}[X]$ is one.

Intuitively, this theorem states that if we observe a single sufficiently long sequence of outcomes from a random process, the cumulative average will settle down to the expected value. Although individual observations fluctuate, their long-term average becomes nearly constant—a concrete manifestation of *stability through repetition*.

The figure below shows this phenomenon for $X_i \sim \text{Exp}(\lambda = 2)$. Each curve represents a running mean trajectory computed from an independent sequence of samples. As n increases, all trajectories approach and remain close to the true mean $\mathbb{E}[X] = \lambda = 2$, illustrating convergence almost surely.

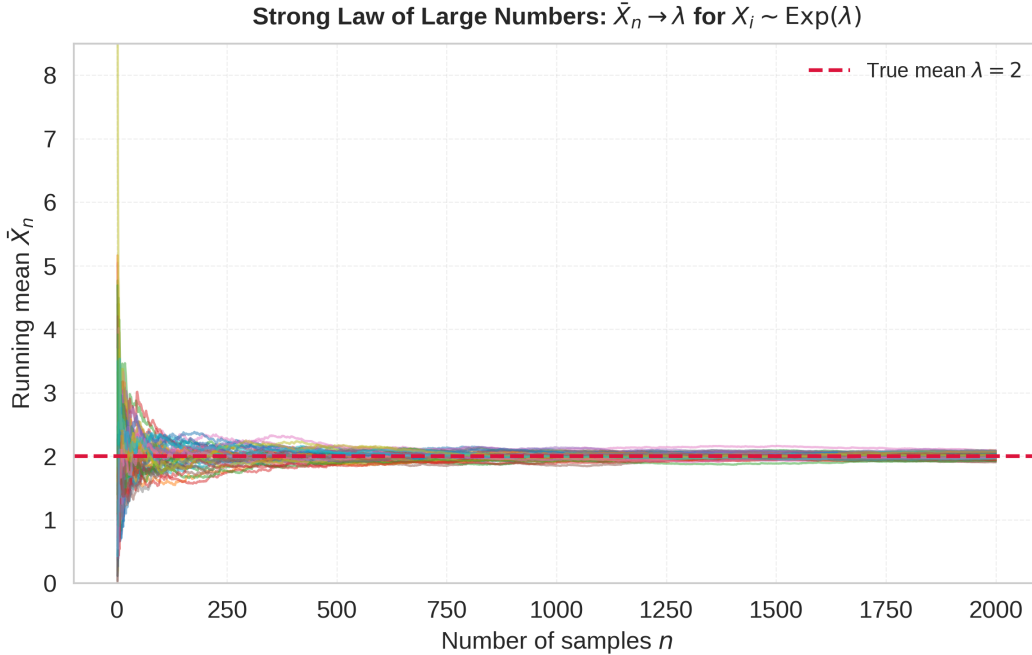


Figure 3: Running mean trajectories $\bar{X}_n$ for 50 independent sequences of $X_i \sim \text{Exp}(\lambda = 2)$. Each curve represents the cumulative average of one sequence. Despite early fluctuations, all trajectories stabilize around the true mean $\mathbb{E}[X] = \lambda = 2$ as n increases, demonstrating almost sure convergence of $\bar{X}_n$ to $\mathbb{E}[X]$.

Difference between WLLN and SLLN. The WLLN and SLLN both assert that the sample mean approaches the true mean as n increases, but they differ in the *mode of convergence*:

- **WLLN:** Convergence *in probability*. The fraction of realizations where $\bar{X}_n$ deviates from $\mathbb{E}[X]$ by more than ϵ becomes negligible as n increases.

$$\forall \epsilon > 0, \quad \mathbb{P}(|\bar{X}_n - \mathbb{E}[X]| > \epsilon) \rightarrow 0.$$

- **SLLN:** Convergence *almost surely*. Every observed realization (sequence) of the process will eventually settle at $\mathbb{E}[X]$, except possibly for a set of measure zero.

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mathbb{E}[X]\right) = 1.$$

In short: the WLLN ensures that “*most samples are close to the truth most of the time*,” while the SLLN guarantees that “*every single sample path eventually converges to the truth*.”

3.3 Central Limit Theorem (CLT)

The Central Limit Theorem provides a probabilistic bridge between an arbitrary distribution and the normal distribution. Let $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} X$ with $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2 < \infty$. Then, as $n \rightarrow \infty$, the standardized sample mean

$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$$

converges in distribution to the standard normal:

$$Z_n \xrightarrow{d} \mathcal{N}(0, 1).$$

This result implies that, regardless of the original shape of the population distribution, the sampling distribution of the standardized mean becomes approximately normal for sufficiently large n . In other words, random variation in averages exhibits a universal Gaussian pattern.

For the Exponential distribution $X_i \sim \text{Exp}(\lambda = 2)$, we have $\mu = \lambda$ and $\sigma = \lambda$. The figure below illustrates the convergence of the sampling distribution of the standardized mean to the standard normal as n increases.

CLT: $Z_n = \sqrt{n} (\bar{X}_n - \mu)/\sigma$ for $X_i \sim \text{Exp}(\lambda)$

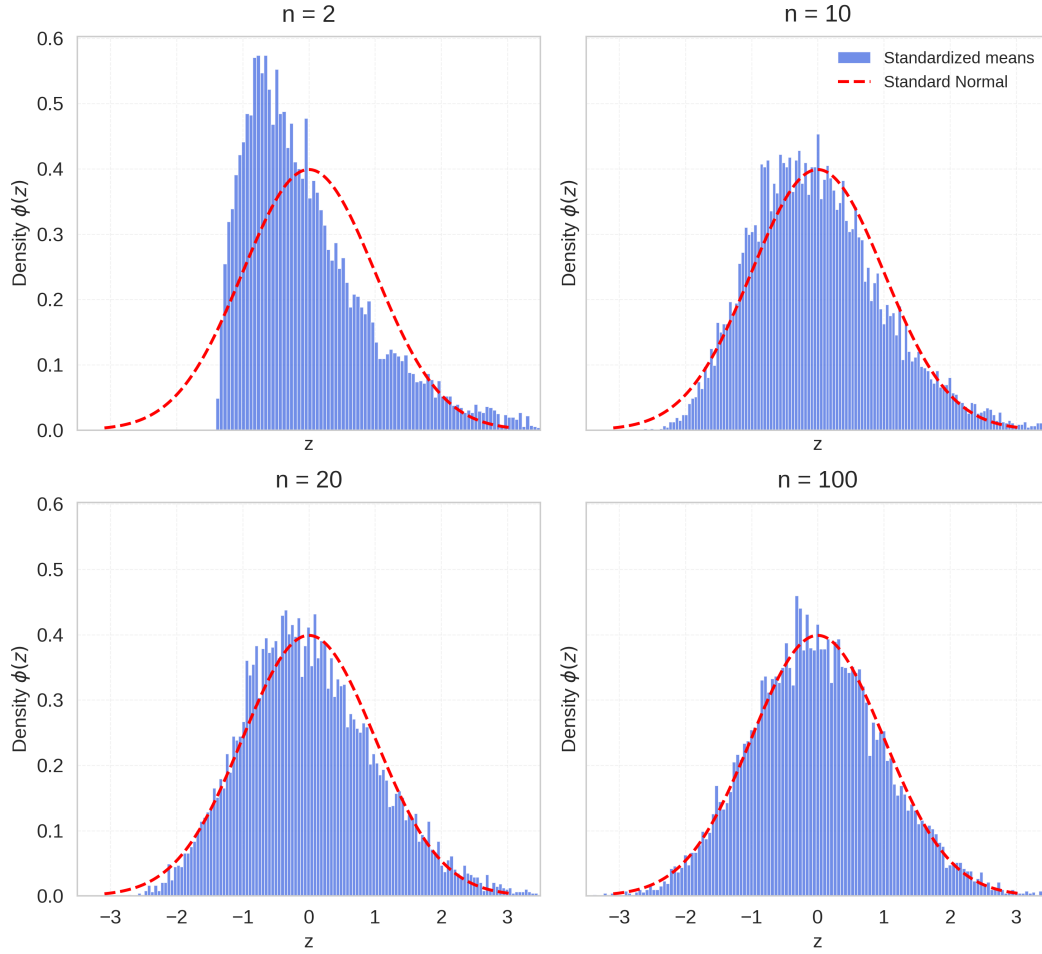


Figure 4: Standardized sampling distributions of $\bar{X}_n$ for $X_i \sim \text{Exp}(\lambda = 2)$. Each histogram shows the distribution of $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ across 1000 trials. As n increases, the distribution becomes more symmetric and approaches $\mathcal{N}(0, 1)$, demonstrating convergence in distribution.

4 Advanced Random Sampling Methods

5 Parameter Estimation

we can estimate the mean of a distribution by replacing Expectation with average. In fact 99.9% of statistics are all about replacing expectations with averages.